

DMV302 Data Mining and Visualizations

‘Cluster Analysis’



Component code and name	DMV302 Data Mining and Visualization
Assignment name	Cluster Analysis
Weighting	35%
Submission deadline	Week 12
Week issued	Week 8

Brief

This assessment requires you to implement a few data mining algorithms for clustering and classification to solve problems in real-world scenarios. You will also need to answer a few questions in this problem set. Some of these questions ask you to explain the design or choice of certain parameters in the algorithms, while others ask you to reflect on the implementation and propose ways to optimize/customize those algorithms.

Problem Statement: Customer Behavior Analysis and Risk Profiling Using Clustering and Classification

A large e-commerce company wants to understand the purchasing behavior of its customers to improve targeted marketing and assess the financial risk associated with different customer segments. You are provided with a dataset containing historical transaction data, demographic details, and past credit/risk indicators.

You are required to:

1. **Cluster** customers based on behavioral and demographic attributes to discover hidden segments.
2. **Classify** new customers into predefined risk categories (e.g., Low, Medium, High Risk) using a supervised learning model.
3. **Interpret** the findings and provide actionable insights.

Please refer to the Task Instructions (below) for details on how to complete this task.

Instructions

Complete the following questions.

Question 1—Clustering (35 %)

- Implement clustering algorithms in Python that reads and clusters data.
 - Preprocess the data: handle missing values, encode categorical data, and normalize. (Note: Drop `Credit_Score` attribute for clustering task)
 - Apply `KMeans` with Euclidean distance and Initiate `K = 3`.
 - Save the source code as `'Clustering.ipynb'`.
 - You were asked to set `K = 3`; however, that choice may not be the most appropriate for the data. Determine the optimal number of clusters using the **Elbow Method** and then validate it using the **Silhouette Score** (*expected high value*).
 - Visualize the clusters in 2D using **PCA** or **t-SNE**.
- Implement hierarchical clustering methods, such as **agglomerative clustering**, to produce cluster outputs.
- Implement the **DBSCAN** algorithm on the same data problem by selecting the suitable value with `MinPoints` and `Eps`.
- Compare the performance of optimized Agglomerative, DBSCAN, and K-means algorithms based on the Dunn Index and Inertia. Save your implementations in the same source file, `'Clustering.ipynb'`.

Table 1. An example of the table

Metrics	K-means	Agglomerative	DBSCAN
Dunn Index			
Inertia			

- Apply the best clustering model on unsupervised Test data, and form the clusters as suggested by your model. Ideally, it is '3' as given below. Calculate count/percentage of:

```
Credit_Score = Good
Credit_Score = Standard
Credit_Score = Poor
```

Question 2—Classification (Supervised Learning) (35 %)

- Split the Train dataset into training and validation sets.
- Use models such as:
 - Decision Trees
 - Random Forest
 - Logistic Regression
 - Support Vector Machine (SVM)
- Evaluate model performance using the Test dataset:
 - Confusion Matrix
 - Accuracy, Precision, Recall, F1 Score
 - ROC-AUC Curve
- Compare models on the test set and select the best one for predicting `Credit_score/Risk_Level`. Save your implementations in the same source file, 'Classification.ipynb'.

Table 2 An example of the table for Question 2.

Model	ACC	REC	F1-Score	PRE

- Model Interpretation using **SHAP values**, **Feature Importance plots**, or **LIME** to interpret the classification model.

Question 3- Prepare a Report on Insights (20%)

- Interpret the clusters — what defines each group behaviorally?
- Which algorithm formed the most well-separated and meaningful clusters?
- Did DBSCAN detect any noise, or outlier points that others missed?
- How do hierarchical clusters differ in structure from K-Means partitions?
- Are some clusters more likely to fall into the High-Risk group, i.e., have the highest proportion of customers with `Credit_Score = Poor`?
- Use visualizations like heatmaps or cluster vs risk bar plots to illustrate overlap.
- Calculate the purity score of each cluster for `Credit_Score=Poor, Good, or Standard`.
- Identify top features that differentiate one cluster from another.
- Use mean feature values per cluster and radar charts or box plots for comparison.
- Which features were most influential in predicting customer risk?
- Are the same features responsible for defining clusters and classifying risk?

General requirements for Python programming

All Python programs in this assessment should comply with the following requirements:

- Follow Python naming conventions. If in doubt, refer to the *Google Python Style Guide* at <https://google.github.io/styleguide/pyguide.html>.
- Be implemented with Python 3 (**NOT** Python 2.x).
- Be implemented with Jupyter Notebook IDE.
- Contain adequate and useful comments.
- Be thoroughly tested and do not contain any syntax or logical errors.

Referencing

Students must use the APA style for citing and referencing research appropriately. Please see more information on referencing at <https://mediadesignschool.libguides.com/apa>.

Submission Guidelines

Naming conventions

DMV302LastNameFirstNameBrief1.zip

- Clustering.ipynb (Jupyter notebook or Google collab link)
- Classification.ipynb
- Report.docx/doc

Submission structure

Submit the code file via the Assessment 2 link in the main navigation menu in DMV302: Assignment 2, by the end of Week 12.

Assessment Criteria

The brief outline of assessment criteria is as follows. For more details on each section, check MyLearn assessment rubrics.

1. **Clustering (35%)**
2. **Classification (35%)**
3. **Report Writing (20%)**
4. **Programming Practices (10%)**

Submission Policy

Submit this task via the **Assessment 2** link in the main navigation menu in DMV302 Data Mining and Visualisation. The Learning Facilitator will provide feedback in the MyLearn portal.

You are required to submit **Python code files (.ipynb)** and **MS-word reports (.doc or .docx)**. Please ensure that you submit individual files inside a single zip and do not zip/compress individual files. Before submitting your assignment, you should check it against the detailed assessment criteria in the following table to ensure that you have satisfactorily addressed all the criteria that will be used to mark your assignment.

Please refer to the Submission Policy on MyLearn for further details.

Academic Integrity

I declare that except where I have referenced, the work I am submitting for this assessment task is my work.

I have read and am aware of Media Design School's Academic Integrity Policy and Procedure viewable online at the <https://www.mediadesignschool.com/policies-and-procedures>.